Grothendieck spectral sequence for compositions of right derived functors

The Stacks Project Authors

stacks-project:derived.tex@a04446e57ec1fbc252a871afcec7752fb2807b14;
GNU FDL 1.2 or later

Editorial note. Original statement, complete author proof, Cartan–Eilenberg construction and both source-local spectral-sequence calculations retained. External original labels remain explicit source pointers, without guessed theorem numbering. Shared pinned my.bib supplied; no collector mathematical repairs.

Allowed standard prerequisites (editorial).

- Abelian categories with enough injectives, bounded-below derived functors and double-complex spectral-sequence notation. Complete source-local Cartan–Eilenberg construction, page computations and composition argument retained. Ordinary injective-resolution and first-quadrant convergence facts remain exactly cited standard prerequisites.

1 Right derived functors and injective resolutions

At this point we can use the material above to define the right derived functors of an additive functor between an abelian category having enough injectives and a general abelian category.

Lemma 1.1. Let $\mathcal{A}$ be an abelian category. Let $I \in \mathrm{Ob}(\mathcal{A})$ be an injective object. Let $I^\bullet$ be a bounded below complex of injectives in $\mathcal{A}$.

1. $I^\bullet$ computes RF relative to $\mathrm{Qis}^+(\mathcal{A})$ for any exact functor $F : K^+(\mathcal{A}) \to \mathcal{D}$ into any triangulated category $\mathcal{D}$.
2. I is right acyclic for any additive functor $F : \mathcal{A} \to \mathcal{B}$ into any abelian category $\mathcal{B}$.

Proof. Part (2) is a direct consequences of part (1) and Definition [original label: `definition-derived-functor`]. To prove (1) let $\alpha : I^\bullet \to K^\bullet$ be a quasi-isomorphism into a complex. By Lemma [original label: `lemma-morphisms-lift`] we see that α has a left inverse. Hence the category $I^\bullet/\mathrm{Qis}^+(\mathcal{A})$ is essentially constant with value $\mathrm{id} : I^\bullet \to I^\bullet$. Thus also the ind-object

$$I^\bullet/\mathrm{Qis}^+(\mathcal{A}) \longrightarrow \mathcal{D}, \quad (I^\bullet \to K^\bullet) \longmapsto F(K^\bullet)$$

is essentially constant with value $F(I^\bullet)$. This proves (1), see Definitions [original label: `definition-right-derived-functor-defined`] and [original label: `definition-computes`]. □

Lemma 1.2. Let $\mathcal{A}$ be an abelian category with enough injectives.

1. For any exact functor $F : K^+(\mathcal{A}) \to \mathcal{D}$ into a triangulated category $\mathcal{D}$ the right derived functor
$$RF : D^+(\mathcal{A}) \longrightarrow \mathcal{D}$$
is everywhere defined.
2. For any additive functor $F : \mathcal{A} \to \mathcal{B}$ into an abelian category $\mathcal{B}$ the right derived functor
$$RF : D^+(\mathcal{A}) \longrightarrow D^+(\mathcal{B})$$
is everywhere defined.

Proof. Combine Lemma 1.1 and Proposition [original label: `proposition-enough-acyclics`] for the second assertion. To see the first assertion combine Lemma [original label: `lemma-injective-resolutions-exist`], Lemma 1.1, Lemma [original label: `lemma-existence-computes`], and Equation ([original label: `equation-everywhere`]). □

Lemma 1.3. Let $\mathcal{A}$ be an abelian category with enough injectives. Let $F : \mathcal{A} \to \mathcal{B}$ be an additive functor.

1. The functor RF is an exact functor $D^+(\mathcal{A}) \to D^+(\mathcal{B})$.
2. The functor RF induces an exact functor $K^+(\mathcal{A}) \to D^+(\mathcal{B})$.
3. The functor RF induces a δ-functor $\mathrm{Comp}^+(\mathcal{A}) \to D^+(\mathcal{B})$.
4. The functor RF induces a δ-functor $\mathcal{A} \to D^+(\mathcal{B})$.

Proof. This lemma simply reviews some of the results obtained so far. Note that by Lemma 1.2 RF is everywhere defined. Here are some references:

1. The derived functor is exact: This boils down to Lemma [original label: `lemma-2-out-of-3-defined`].
2. This is true because $K^+(\mathcal{A}) \to D^+(\mathcal{A})$ is exact and compositions of exact functors are exact.
3. This is true because $\mathrm{Comp}^+(\mathcal{A}) \to D^+(\mathcal{A})$ is a δ-functor, see Lemma [original label: `lemma-derived-canonical-delta-functor`].
4. This is true because $\mathcal{A} \to \mathrm{Comp}^+(\mathcal{A})$ is exact and precomposing a δ-functor by an exact functor gives a δ-functor.

□

Lemma 1.4. Let $\mathcal{A}$ be an abelian category with enough injectives. Let $F : \mathcal{A} \to \mathcal{B}$ be a left exact functor.

1. For any short exact sequence $0 \to A^\bullet \to B^\bullet \to C^\bullet \to 0$ of complexes in $\text{Comp}^+(\mathcal{A})$ there is an associated long exact sequence

$$\ldots \to H^i(RF(A^\bullet)) \to H^i(RF(B^\bullet)) \to H^i(RF(C^\bullet)) \to H^{i+1}(RF(A^\bullet)) \to \ldots$$

2. The functors $R^iF : \mathcal{A} \to \mathcal{B}$ are zero for $i < 0$. Also $R^0F = F : \mathcal{A} \to \mathcal{B}$.

3. We have $R^iF(I) = 0$ for $i > 0$ and I injective.

4. The sequence (R^iF, δ) forms a universal δ-functor (see Homology, Definition [original label: *homology-definition-universal-delta-functor*]) from $\mathcal{A}$ to $\mathcal{B}$.

Proof. This lemma simply reviews some of the results obtained so far. Note that by Lemma 1.2 RF is everywhere defined. Here are some references:

1. This follows from Lemma 1.3 part (3) combined with the long exact cohomology sequence ([original label: `equation-long-exact-cohomology-sequence-D`]) for $D^+(\mathcal{B})$.

2. This is Lemma [original label: `lemma-left-exact-higher-derived`].

3. This is the fact that injective objects are acyclic.

4. This is Lemma [original label: `lemma-right-derived-delta-functor`].

□

2 Cartan-Eilenberg resolutions

This section can be expanded. The material can be generalized and applied in more cases. Resolutions need not use injectives and the method also works in the unbounded case in some situations.

Definition 2.1. Let $\mathcal{A}$ be an abelian category. Let $K^\bullet$ be a bounded below complex. A Cartan-Eilenberg resolution of $K^\bullet$ is given by a double complex $I^{\bullet,\bullet}$ and a morphism of complexes $\epsilon : K^\bullet \to I^{\bullet,0}$ with the following properties:

1. There exists a $i \ll 0$ such that $I^{p,q} = 0$ for all $p < i$ and all q.

2. We have $I^{p,q} = 0$ if $q < 0$.

3. The complex $I^{p,\bullet}$ is an injective resolution of K^p.

4. The complex $\text{Ker}(d_1^{p,\bullet})$ is an injective resolution of $\text{Ker}(d_K^p)$.

5. The complex $\text{Im}(d_1^{p,\bullet})$ is an injective resolution of $\text{Im}(d_K^p)$.

6. The complex $H_I^p(I^{\bullet,\bullet})$ is an injective resolution of $H^p(K^\bullet)$.

Lemma 2.2. Let $\mathcal{A}$ be an abelian category with enough injectives. Let $K^\bullet$ be a bounded below complex. There exists a Cartan-Eilenberg resolution of $K^\bullet$.

Proof. Suppose that $K^p = 0$ for $p < n$. Decompose $K^\bullet$ into short exact sequences as follows: Set $Z^p = \mathrm{Ker}(d^p)$, $B^p = \mathrm{Im}(d^{p-1})$, $H^p = Z^p/B^p$, and consider

$$\begin{aligned} 0 \to Z^n \to K^n \to B^{n+1} \to 0 \\ 0 \to B^{n+1} \to Z^{n+1} \to H^{n+1} \to 0 \\ 0 \to Z^{n+1} \to K^{n+1} \to B^{n+2} \to 0 \\ 0 \to B^{n+2} \to Z^{n+2} \to H^{n+2} \to 0 \\ \ldots \end{aligned}$$

Set $I^{p,q} = 0$ for $p < n$. Inductively we choose injective resolutions as follows:

1. Choose an injective resolution $Z^n \to J_Z^{n,\bullet}$ with $J_Z^{n,m} = 0$ for $m < 0$, see Lemma [original label: `lemma-injective-resolutions-exist`].
2. Using Lemma [original label: `lemma-injective-resolution-ses`] choose injective resolutions $K^n \to I^{n,\bullet}$, $B^{n+1} \to J_B^{n+1,\bullet}$ with $I^{n,m} = 0$ for $m < 0$ and $J_B^{n+1,m} = 0$ for $m < 0$, and an exact sequence of complexes $0 \to J_Z^{n,\bullet} \to I^{n,\bullet} \to J_B^{n+1,\bullet} \to 0$ compatible with the short exact sequence $0 \to Z^n \to K^n \to B^{n+1} \to 0$.
3. Using Lemma [original label: `lemma-injective-resolution-ses`] choose injective resolutions $Z^{n+1} \to J_Z^{n+1,\bullet}$, $H^{n+1} \to J_H^{n+1,\bullet}$ with $J_Z^{n+1,m} = 0$ for $m < 0$ and $J_H^{n+1,m} = 0$ for $m < 0$ and an exact sequence of complexes $0 \to J_B^{n+1,\bullet} \to J_Z^{n+1,\bullet} \to J_H^{n+1,\bullet} \to 0$ compatible with the short exact sequence $0 \to B^{n+1} \to Z^{n+1} \to H^{n+1} \to 0$.
4. Etc.

Taking as maps $d_1^\bullet : I^{p,\bullet} \to I^{p+1,\bullet}$ the compositions $I^{p,\bullet} \to J_B^{p+1,\bullet} \to J_Z^{p+1,\bullet} \to I^{p+1,\bullet}$ everything is clear. □

Lemma 2.3. Let $F : \mathcal{A} \to \mathcal{B}$ be a left exact functor of abelian categories. Let $K^\bullet$ be a bounded below complex of $\mathcal{A}$. Let $I^{\bullet,\bullet}$ be a Cartan-Eilenberg resolution for $K^\bullet$. The spectral sequences $({}'E_r, {}'d_r)_{r \geq 0}$ and $({}''E_r, {}''d_r)_{r \geq 0}$ associated to the double complex $F(I^{\bullet,\bullet})$ satisfy the relations

$${}'E_1^{p,q} = R^q F(K^p) \quad \text{and} \quad {}''E_2^{p,q} = R^p F(H^q(K^\bullet))$$

Moreover, these spectral sequences are bounded, converge to $H^*(RF(K^\bullet))$, and the associated induced filtrations on $H^n(RF(K^\bullet))$ are finite.

Proof. We will use the following remarks without further mention:

1. As $I^{p,\bullet}$ is an injective resolution of K^p we see that RF is defined at $K^p[0]$ with value $F(I^{p,\bullet})$.
2. As $H_I^p(I^{\bullet,\bullet})$ is an injective resolution of $H^p(K^\bullet)$ the derived functor RF is defined at $H^p(K^\bullet)[0]$ with value $F(H_I^p(I^{\bullet,\bullet}))$.
3. By Homology, Lemma [original label: `homology-lemma-double-complex-gives-resolution`] the total complex $\mathrm{Tot}(I^{\bullet,\bullet})$ is an injective resolution of $K^\bullet$. Hence RF is defined at $K^\bullet$ with value $F(\mathrm{Tot}(I^{\bullet,\bullet}))$.

Consider the two spectral sequences associated to the double complex $L^{\bullet,\bullet} = F(I^{\bullet,\bullet})$, see Homology, Lemma [original label: `homology-lemma-ss-double-complex`]. These are both bounded, converge to $H^*(\mathrm{Tot}(L^{\bullet,\bullet}))$, and induce finite filtrations on $H^n(\mathrm{Tot}(L^{\bullet,\bullet}))$, see Homology, Lemma [original label: `homology-lemma-first-quadrant-ss`]. Since $\mathrm{Tot}(L^{\bullet,\bullet}) = \mathrm{Tot}(F(I^{\bullet,\bullet})) = F(\mathrm{Tot}(I^{\bullet,\bullet}))$ computes $RF(K^\bullet)$ we find the final assertion of the lemma holds true.

Computation of the first spectral sequence. We have ${}'E_1^{p,q} = H^q(L^{p,\bullet})$ in other words

$$ {}'E_1^{p,q} = H^q(F(I^{p,\bullet})) = R^qF(K^p) $$

as desired. Observe for later use that the maps ${}'d_1^{p,q} : {}'E_1^{p,q} \to {}'E_1^{p+1,q}$ are the maps $R^qF(K^p) \to R^qF(K^{p+1})$ induced by $K^p \to K^{p+1}$ and the fact that R^qF is a functor.

Computation of the second spectral sequence. We have ${}''E_1^{p,q} = H^q(L^{\bullet,p}) = H^q(F(I^{\bullet,p}))$. Note that the complex $I^{\bullet,p}$ is bounded below, consists of injectives, and moreover each kernel, image, and cohomology group of the differentials is an injective object of $\mathcal{A}$. Hence we can split the differentials, i.e., each differential is a split surjection onto a direct summand. It follows that the same is true after applying F. Hence ${}''E_1^{p,q} = F(H^q(I^{\bullet,p})) = F(H_I^q(I^{\bullet,p}))$. The differentials on this are $(-1)^q$ times F applied to the differential of the complex $H_I^p(I^{\bullet,\bullet})$ which is an injective resolution of $H^p(K^\bullet)$. Hence the description of the E_2 terms. $\square$

Remark 2.4. The spectral sequences of Lemma 2.3 are functorial in the complex $K^\bullet$. This follows from functoriality properties of Cartan-Eilenberg resolutions. On the other hand, they are both examples of a more general spectral sequence which may be associated to a filtered complex of $\mathcal{A}$. The functoriality will follow from its construction. We will return to this in the section on the filtered derived category, see Remark [original label: `remark-final-functorial`].

3 Composition of right derived functors

Sometimes we can compute the right derived functor of a composition. Suppose that $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories. Let $F : \mathcal{A} \to \mathcal{B}$ and $G : \mathcal{B} \to \mathcal{C}$ be left exact functors. Assume that the right derived functors $RF : D^+(\mathcal{A}) \to D^+(\mathcal{B})$, $RG : D^+(\mathcal{B}) \to D^+(\mathcal{C})$, and $R(G \circ F) : D^+(\mathcal{A}) \to D^+(\mathcal{C})$ are everywhere defined. Then there exists a canonical transformation

$$ t : R(G \circ F) \longrightarrow RG \circ RF $$

of functors from $D^+(\mathcal{A})$ to $D^+(\mathcal{C})$, see Lemma [original label: `lemma-compose-derived-functors-general`]. This transformation need not always be an isomorphism.

Lemma 3.1. Let $\mathcal{A}, \mathcal{B}, \mathcal{C}$ be abelian categories. Let $F : \mathcal{A} \to \mathcal{B}$ and $G : \mathcal{B} \to \mathcal{C}$ be left exact functors. Assume $\mathcal{A}$, $\mathcal{B}$ have enough injectives. The following are equivalent

1. $F(I)$ is right acyclic for G for each injective object I of $\mathcal{A}$, and
2. the canonical map
$$ t : R(G \circ F) \longrightarrow RG \circ RF. $$
is isomorphism of functors from $D^+(\mathcal{A})$ to $D^+(\mathcal{C})$.

Proof. If (2) holds, then (1) follows by evaluating the isomorphism t on $RF(I) = F(I)$. Conversely, assume (1) holds. Let $A^\bullet$ be a bounded below complex of $\mathcal{A}$. Choose an injective resolution $A^\bullet \to I^\bullet$. The map t is given (see proof of Lemma [original label: `lemma-compose-derived-functors-general`]) by the maps

$$R(G \circ F)(A^\bullet) = (G \circ F)(I^\bullet) = G(F(I^\bullet))) \to RG(F(I^\bullet)) = RG(RF(A^\bullet))$$

where the arrow is an isomorphism by Lemma [original label: `lemma-leray-acyclicity`]. □

Lemma 3.2 (Grothendieck spectral sequence). With assumptions as in Lemma 3.1 and assuming the equivalent conditions (1) and (2) hold. Let X be an object of $D^+(\mathcal{A})$. There exists a spectral sequence $(E_r, d_r)_{r \geq 0}$ consisting of bigraded objects E_r of $\mathcal{C}$ with d_r of bidegree $(r, -r+1)$ and with

$$E_2^{p,q} = R^p G(H^q(RF(X)))$$

Moreover, this spectral sequence is bounded, converges to $H^*(R(G \circ F)(X))$, and induces a finite filtration on each $H^n(R(G \circ F)(X))$.

For an object A of $\mathcal{A}$ we get $E_2^{p,q} = R^p G(R^q F(A))$ converging to $R^{p+q}(G \circ F)(A)$.

Proof. We may represent X by a bounded below complex $A^\bullet$. Choose an injective resolution $A^\bullet \to I^\bullet$. Choose a Cartan-Eilenberg resolution $F(I^\bullet) \to I^{\bullet,\bullet}$ using Lemma 2.2. Apply the second spectral sequence of Lemma 2.3. □